Allen Tu

Ph.D. Student, Computer Science — University of Maryland, College Park
3D/4D Digital Twins, World Models, Generative AI

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Selected Publications

1. **Allen Tu***, H. Ying*, A. Hanson, Y. Lee, T. Goldstein, and M. Zwicker, 'SpeeDe3DGS: Speedy Deformable 3D Gaussian Splatting with Temporal Pruning and Motion Grouping'. *CVPR*, 2026.
2. P. Asthana, A. Hanson, **Allen Tu**, T. Goldstein, M. Zwicker, and A. Varshney, 'SplatSuRe: Selective Super-Resolution for Multi-view Consistent 3D Gaussian Splatting'. *CVPR*, 2026.
3. **Allen Tu**, K. Narayan, J. Gleason, J. Xu, M. Meyn, and V. Patel, 'TransFIRA: Transfer Learning for Face Image Recognizability Assessment'. *FG*, 2026 (**Oral Presentation; Invited to submit to IEEE T-BIOM Special Issue**).
4. A. Hanson*, **Allen Tu***, V. Singla, M. Jayawardhana, M. Zwicker, and T. Goldstein, 'PUP 3D-GS: Principled Uncertainty Pruning for 3D Gaussian Splatting'. *CVPR*, 2025.
5. A. Hanson, **Allen Tu**, G. Lin, V. Singla, M. Zwicker, and T. Goldstein, 'Speedy-Splat: Fast 3D Gaussian Splatting with Sparse Pixels and Sparse Primitives'. *CVPR*, 2025.

* denotes equal contribution.

Education

University of Maryland, College Park — College Park, MD
Ph.D. Student, Computer Science — *January 2025 – May 2028 (Expected)*
- **Advisors:** Professor Tom Goldstein, Professor Matthias Zwicker

B.S./M.S. in Computer Science, Minor in Statistics — *August 2019 – December 2024*

Research Experience

Amazon — Bellevue, WA
Applied Science Intern — *June 2026 – Present*
- Researching 3D digital twins, world models, and agentic pipelines for physical AI simulation in the Last Mile Delivery and Automation org
- Created a method that reconstructs simulation-ready 3D digital twins directly from fisheye imagery, combining photorealistic Gaussian Splatting over the camera's full field of view with an aligned mesh for collisions
- Constructing scalable pipelines that produce photorealistic, metric-scale 3D digital twin simulation environments from large-scale passive data collection

University of Maryland Institute of Advanced Computer Studies — College Park, MD
Graduate Research Assistant — *August 2023 – Present*
- Currently researching latent world models, unconstrained 3D/4D reconstruction, and generative modeling under Professors Tom Goldstein and Matthias Zwicker
- Published papers on efficient 3D/4D Gaussian Splatting [1, 4, 5], 3D reconstruction from degraded and fisheye imagery [2], conditioned CAD model generation, and biometric recognition [3]
- Developed scalable 3D reconstruction pipelines deployed for unconstrained novel-view synthesis in real-world environments under the IARPA Walk-through Rendering from Images of Varying Altitude (WRIVA) program

Systems & Technology Research — Arlington, VA
Computer Vision Research Intern — *June 2022 – January 2026*
- Engineered multimodal biometric systems fusing face, body, and gait recognition for robust identification under severe conditions in the IARPA Biometric Recognition and Identification at Altitude and Range (BRIAR) program
- Introduced an explainable transfer-learning method for face image quality assessment (FIQA), enabling recognizability-aware probe filtering and weighted template aggregation for face and body recognition [3]

Selected Honors & Service

FG 2026 Doctoral Consortium

Workshop Organizer: SPAR-3D: Security, Privacy, & Adversarial Robustness in 3D Generative Vision Models, CVPR 2026